

Module 5 — Vehicle & Transportation Management

(modules/vehicles)

Purpose

Track all vehicle lifecycle costs (fuel, maintenance, insurance, taxes, inspections), compute fuel efficiency and cost-per-km, support multi-payer expense splitting, and provide per-member accountability and total cost-of-ownership analytics. Integrate Portuguese compliance requirements (IUC annual tax, IPO mandatory inspection) and support mixed fuel fleets (petrol, diesel, LPG, electric).

Actors & User Stories

- *As a driver*, I log fuel purchases with odometer, and the system computes L/100km automatically.
- *As a household manager*, I track multi-vehicle expenses and split costs by driver (e.g., User A: 60%, User B: 40%).
- *As a tax/compliance officer*, I receive reminders for IUC renewal and IPO inspection due dates.
- *As an EV owner*, I log home and public charging separately and track cost-per-kWh.
- *As an analyst*, I view fuel efficiency trends, cost-per-km evolution, and per-member cost accountability.

Data Model

All monetary fields (`_eur` suffix) are **NUMERIC(10,2)** decimal EUR amounts (e.g. **19.99**), never minor-unit integers; UI displays them pt-PT-formatted (e.g. **19,99€**).

- **Vehicle**: `id`, `entity_id`, `name`, `make`, `model`, `plate_masked`, `fuel_type`(PETROL_95|PETROL_98|DIESEL|LPG|ELECTRIC), `odometer_km`, `purchase_date`, `purchase_price_eur`, `resale_value_snapshot_id?`, `is_deleted`, `tags[]`.
- **VehicleExpense** (base): `id`, `vehicle_id`, `expense_type`(FUEL|MAINTENANCE|INSURANCE|TAX|REPAIR|OTHER), `date`, `amount_eur`, `odometer_km`, `description`, `document_id`, `transaction_id?`, `payer_split` JSONB, `tags[]`, `confidence`, `decision_reasons`.
- **FuelEvent** (efficiency analytics): `id`, `vehicle_id`, `date`, `odometer_km`, `fuel_type`, `liters` NUMERIC(8,3), `price_per_liter_eur`, `total_cost_eur`, `station_merchant_id`, `is_full_fill` BOOLEAN, `source_expense_id`, `confidence`.
L/100km computed only between consecutive full fills; partial fills recorded but excluded.
- **MaintenanceSchedule**: `id`, `vehicle_id`, `service_type`(OIL|FILTER|TIRES|BRAKES|INSPECTION|OTHER), `interval_km`, `interval_months`, `next_due_km`, `next_due_date`, `notes`, `reminder_sent_at`.
- **RecurringVehicleCost**: `id`, `vehicle_id`, `cost_type`(INSURANCE|IUC|PORTAGEM|INSPECTION|REGISTRATION), `amount_eur`, `cadence`(MONTHLY|QUARTERLY|ANNUAL), `due_date`, `last_paid_date`, `next_due_date`, `notes`, `linked_transaction_id?`.
- **VehicleInsurance**: `id`, `vehicle_id`, `insurer_name`, `policy_number`, `premium_amount_eur`, `coverage_type`, `renewal_date`, `document_id`.

- **TireLog:** `id, vehicle_id, brand, size, install_date, install_odometer_km, removal_date, removal_odometer_km, cost_eur, notes`.
- **EVChargeEvent:** `id, vehicle_id, date, odometer_km, energy_kwh NUMERIC(7,2), cost_eur, charger_type(HOME|PUBLIC_DC|PUBLIC_AC), location, cost_per_kwh_eur`.
- **VehicleResaleSnapshot:** `id, vehicle_id, as_of_date, estimated_value_eur, source_url, notes`.
- **Derived:** `l_per_100km` (inter-full-fill), `cost_per_km_eur`, `cost_per_month_eur`, `total_ownership_cost_eur`. Money as `amount_eur NUMERIC(10,2)`; `payer_split JSONB [{entity_id, percentage}]` summing exactly **100**.

Functional Requirements

- **FR-5.1 Expense Logging.** Record fuel, maintenance, insurance, taxes, repairs; auto-linked to bank transactions by date/amount/merchant.
- **FR-5.2 Fuel Event Tracking & Efficiency.** Log with odometer, liters (full/partial), date. Compute `L/100km` only between consecutive full fills; partial fills excluded and clearly marked.
- **FR-5.3 Multi-Payer Cost Splitting.** `payer_split` array; validate sum = 100; per-member rollup.
- **FR-5.4 Mileage Consistency Validation.** Odometer strictly monotonic per vehicle; retroactive entries with earlier odometer rejected.
- **FR-5.5 Maintenance Scheduling & Reminders.** Intervals (km/months); auto-generate reminder tasks in Review Queue when due.
- **FR-5.6 Recurring Cost Management.** Insurance, IUC, portagens, inspection; auto-reconcile with bank; alert on missed/changed amounts.
- **FR-5.7 Portuguese Compliance Tracking.** IUC (annual road tax; 30-day reminder), IPO (biennial inspection; due-date + pass/fail), Via Verde / Portagens (toll charges; stored-value + reconciliation).
- **FR-5.8 EV Charging Cost Tracking.** Log home/public charging (kWh + cost); configurable home rate (€/kWh).
- **FR-5.9 Resale Value Snapshots.** Link to Assets; periodically snapshot estimated resale value.
- **FR-5.10 Trip-Level Tagging.** Tag expenses/events (`#commute`, `#vacation`, `#work`, `#errands`).
- **FR-5.11 Tire & Oil-Change Tracking.** Dedicated tire log + oil-change intervals for proactive maintenance.

Automation Rules

- Efficiency computed only between full fills: $(\text{liters_consumed} / \text{distance_km}) \times 100$; partial fills break the chain.
- Fuel events auto-matched to bank transactions (± 2 days, amount $\pm 5\%$, merchant similarity).
- Recurring costs auto-suggested to matching transactions; confirmations advance `last_paid_date`.
- Maintenance reminders when `next_due_km ≤ odometer` or `next_due_date ≤ today`; creates ReviewTask; completion updates next interval.
- IUC/IPO due-date alerts 30 days prior (configurable).
- Per-member cost = $\Sigma (\text{expense_amount} \times \text{payer_split}[\text{member}].\text{percentage} / 100)$.

UI / Screens

- **Vehicle Manager:** list/add/edit (make, model, plate, fuel type, purchase price, resale link).
- **Fuel Log:** date, odometer, liters, price/L, `is_full_fill` (highlighted if partial), linked transaction.

- **Expense Log:** type-filtered, odometer, amount, split summary, tags.
- **Efficiency Dashboard:** L/100km trend (excludes partials); cost-per-km; annual summary.
- **Maintenance Board:** scheduled services, completion, next-due countdown.
- **Recurring Cost Calendar:** IUC/IPO/portages timeline, reconciliation status.
- **Cost-Split Summary:** per-member accountability; total ownership cost; cost-per-km per member.
- **EV Charging Dashboard:** kWh vs cost, home vs public split, estimated annual cost.

API Surface

CRUD /vehicles · CRUD /fuel-events · CRUD /vehicle-expenses · CRUD /maintenance-schedules · CRUD /recurring-costs · POST /vehicles/{id}/confirm-expense · GET /vehicles/{id}/analytics · GET /vehicles/{id}/maintenance-forecast · GET /vehicles/iuc-ipo-calendar · POST /ev-charge-events · GET /vehicles/{id}/resale-snapshots.

Analytics & KPIs

- Fuel efficiency trend (L/100km per vehicle/fuel type, rolling 3-month avg).
- Cost metrics (cost-per-km, cost-per-month, total cost of ownership).
- Per-member attribution (cost share, by trip tag).
- Compliance readiness (IUC/IPO due dates, overdue flags).
- Maintenance forecast (upcoming services + estimated cost).
- EV economics (cost-per-kWh home vs public; annual electricity vs petrol/diesel equivalent).

Edge Cases & Validation

- **Odometer monotonicity:** retroactive event with earlier odometer → error.
- **Efficiency calc:** only between full fills; zero liters/distance → null (flagged "incomplete").
- **Payer split:** sum exactly 100; fractional allowed (33.33/33.33/33.34); rejected at API if invalid.
- **Partial fill:** marked and excluded from efficiency buckets; user educated on impact.
- **Multi-fuel vehicles:** L/100km not computed across fuel-type boundaries.
- **Recurring cost overrun:** actual >10% over expected → anomaly; confirm or update expected.
- **Resale value expiry:** snapshots >12 months grayed out; annual refresh encouraged.

Additional / Enriched Requirements

1. **Portuguese IUC Compliance:** automate due-date tracking; optional CO₂-emission-based tax lookup; 30-day-prior reminder.
2. **Multi-Fuel Fleet Efficiency:** separate L/100km per fuel type; side-by-side comparison.
3. **Trip-Level Cost Attribution:** tag by trip; aggregate cost-per-trip insights.
4. **Tire Service Integration:** rotations, wear patterns, replacement intervals; cost-per-tire-life amortization.
5. **EV Home Charging Rate Config:** household €/kWh per tariff period; auto-compute home cost; compare to public.
6. **Resale Value Linkage:** Vehicle → Assets link; depreciation curve.
7. **Via Verde Integration Hint:** PORTAGEM cost type → toll pass balance/consumption affordance.
8. **Anomaly Detection:** efficiency drop >15% YoY (engine issue); fuel price/L spike (market vs rogue entry).

9. **Commute/Work Mileage Tracking:** tag commute (potentially deductible); business vs personal analytics.

Open Questions / Decisions

1. **EV home charging rate global or per-vehicle?** → *Per-household with per-vehicle override if multi-rate tariff.*
2. **L/100km rounding?** → *2 decimals.*
3. **Partial-fill recovery?** → *No auto-repair; user toggles `is_full_fill` to reconcile.*
4. **IUC auto-renewal suggestion?** → *Yes, with confirmation.*
5. **Trip attribution with multi-payer split?** → *Yes; $\text{trip cost} = \text{expense} \times \text{payer_split} \%$.*

Definition of Done

- All FRs + edge cases implemented; ≥80% test coverage on domain logic.
- Odometer monotonicity enforced by DB constraint + validation.
- Payer split sum-to-100 enforced at API + DB; fractional supported.
- L/100km correct between full fills; partials excluded.
- Maintenance reminders routed to Review Queue; completion updates schedule.
- IUC/IPO reminders fire 30 days prior; due-date calendar rendered.
- Recurring cost reconciliation; anomaly alerts on amount change.
- EV charging events logged; cost-per-kWh metrics.
- Per-member rollup queries <500ms over 10 years.
- UI screens: manager, fuel log, expense log, efficiency dashboard, cost-split, maintenance board, compliance calendar.

Integration Contract

- **Exposes:** `Vehicle`, `VehicleExpense`, `FuelEvent` as link targets for Banking (M2); per-member cost series feed Dashboards and Household analytics.
- **Consumes:** `Entity` (payer attribution), `Merchant` (stations/garages/insurers), `Category`, `Transaction` (M2), `Document`, `Asset` (resale snapshot).
- **Automation Hook:** Recurring cost matching in Reconciliation Engine (M2); Maintenance reminders for Review Queue; anomaly detection on efficiency/charge amounts.